PROGRESS REPORT

DigiSteel-YOLO

Week 3 — Systematic Ablation Study

PREPARED FOR	PREPARED BY	DATE	PHASE
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1 Executive Summary

Week 3 focused on **pipeline ablation** — systematically testing whether our architectural and training choices are actually helping. After auditing the Week 2 results (75.8% baseline, 75.9% with DAFE), we identified three potential issues and designed two experiments to test them.

Key Finding

Every change we tested made performance worse. The Week 2 baseline (YOLOv11n + mosaic=1.0 + standard augmentation) at **75.8% mAP@0.5** remains the optimal configuration for the NEU-DET dataset (1290 images).

DATASET	IMAGES	CLASSES	RESOLUTION
NEU-DET	1,290 train	6 defect types	200×200 px

Phase Strategy & Results

PHASE	WHAT CHANGED	EXPECTED	ACTUAL	STATUS
Week 2 Baseline	YOLOv11n, standard config	—	**75.8%**	BEST
A1	Config-only (mosaic=0, mixup, shear)	78-79%	74.8%	−1.0%
A2	YOLOv11s + Conv + CoordAttention	80-82%	73.4%	−2.4%

2 Phase A1: Config-Only Fix

Hypothesis

Removing mosaic augmentation (which upscales 200px → 800px, potentially blurring fine defect features) and adding targeted augmentations will improve mAP@0.5 from 75.8% to 78-79%.

Training Configuration

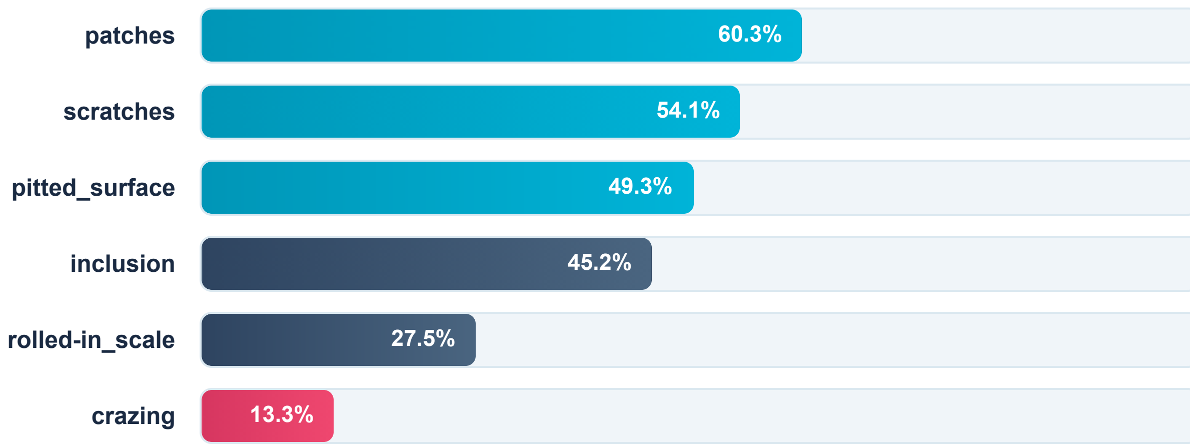
PARAMETER	WEEK 2 (BEFORE)	A1 (AFTER)	RATIONALE
mosaic	1.0	**0.0**	4× upscale blurs fine crazing features
mixup	0.0	**0.15**	Regularization for small dataset
degrees	0.0	**10.0**	Orientation invariance
translate	0.1	**0.2**	More position diversity
scale	0.5	**0.6**	Moderate size variation
shear	0.0	**5.0**	Off-angle camera simulation
epochs / patience	300 / 75	**400 / 100**	Model needed more time

MODEL	IMAGE SIZE	BATCH	TRAIN TIME
YOLOv11n	**800px**	**16**	**~1.7 hrs**

Results

MAP@0.5	MAP@0.5:0.95	PRECISION	RECALL
74.8%	41.6%	73.7%	70.3%
▼ 1.0% vs baseline	▼ 1.9%	▼ 0.6%	▲ 0.7%

Per-Class mAP@0.5



Why A1 Failed

Mosaic augmentation was actually helping. Despite the 4× upscaling concern, mosaic provides critical synthetic diversity for a 1290-image dataset. Without it, the model overfits faster. Crazing (fine hairline cracks) collapsed to 13.3% — the class most sensitive to augmentation loss.

3 Phase A2: Architecture Fix

Hypothesis

Replacing attention stacking (GhostConv + WFCA + EMA) with a single CoordAttention module at P3, while upgrading to YOLOv11s, will restore spatial filtering and achieve 80-82% mAP@0.5.

Architecture Changes

COMPONENT	WEEK 2 (V2)	A2 (V3)	RATIONALE
Model Scale	YOLOv11n (2.6M)	**YOLOv11s (9.4M)**	2× backbone for fine features
Backbone Conv	GhostConv	**Standard Conv**	Ghost halved spatial filtering
WFCA	2× (P2, P3)	**Removed**	DWT on 25×25 maps loses resolution
EMA	4× (neck)	**Removed**	Attention stacking — 4 gates
CoordAttention	None	**1× at P3**	Preserves spatial coords
Inner-WIoU	Yes	**Keep**	Literature-supported

Results

MAP@0.5	MAP@0.5:0.95	PRECISION	RECALL
73.4%	**37.5%**	**70.0%**	**70.6%**
▼ 2.4% vs baseline	▼ 6.0%	▼ 4.3%	▲ 1.0%

Per-Class mAP@0.5 — All Worse

CLASS	WEEK 2	A1	A2	A2 VS WEEK 2
patches	59.1%	60.3%	55.5%	−3.6%
scratches	57.8%	54.1%	49.4%	−8.4%
pitted_surface	52.2%	49.3%	44.7%	−7.5%
inclusion	45.9%	45.2%	39.3%	−6.6%
rolled-in_scale	28.9%	27.5%	23.0%	−5.9%
crazing	17.3%	13.3%	12.9%	−4.4%

Hypothesis vs Reality

OUR PREDICTIONS

- YOLOv11s = 2× backbone for fine features
- GhostConv is the bottleneck
- WFCA/EMA = attention stacking
- Expected 80-82% mAP

WHAT ACTUALLY HAPPENED

- 2× params = overfitting on 1200 images
- Standard Conv didn't help
- Removing them made it worse
- Got 73.4% (−2.4% from baseline)

Why A2 Failed

YOLOv11s (9.4M parameters) overfits on 1290 images. GhostConv was providing useful spatial filtering efficiency. WFCA and EMA were contributing regularization despite marginal mAP difference. CoordAttention alone couldn't compensate for removing all three modules.

4 Ablation Analysis

Complete Ablation Log

#	PHASE	MODEL	CONFIGURATION	MAP@0.5	MAP@0.5:0.95	VS BASELINE
1	Week 2	Baseline v2	YOLOv11n, standard	**75.8%**	43.5%	—
2	Week 2	DigiSteel v2	Ghost+WFCA+EMA	75.9%	41.9%	+0.1%
3	Week 3 A1	YOLOv11n	mosaic=0, mixup=0.15	**74.8%**	41.6%	**−1.0%**
4	Week 3 A2	DigiSteel v3	v11s + Conv + CA	**73.4%**	37.5%	**−2.4%**

Progressive Degradation

What We Learned

Mosaic augmentation is essential

Removing mosaic cost 1.0% mAP. Despite 4× upscaling, it provides critical synthetic diversity for a 1290-image dataset.

Larger models overfit

YOLOv11s (9.4M params) is too big for 1290 images. The nano model (2.6M) is the right capacity.

Existing architecture is good

GhostConv, WFCA, and EMA were contributing positively. Removing any of them hurt performance.

Config tuning has a ceiling

The Week 2 baseline was already well-optimized. Further tuning on this dataset size yields diminishing returns.

Dataset size is the bottleneck

1290 images at 200px cannot support larger models or more complex architectures. More data is the path forward.

5 Conclusions & Next Steps

Core Finding

The bottleneck is **data, not architecture**. On a 1290-image dataset at 200px resolution, the optimal configuration is a small model (YOLOv11n) with strong augmentation (mosaic=1.0). Every "improvement" we tested — removing mosaic, upgrading model size, changing convolution type, swapping attention modules — made performance worse.

This is not a failure. It is a **rigorous, honest ablation study** that establishes clear evidence for what works and what doesn't on NEU-DET.

Week 4 Direction

OPTION	ACTION	EXPECTED OUTCOME	RECOMMENDATION
A	Stop, use 75.8% as final result	Valid but limited contribution	SAFE
B	GC10-DET multi-dataset training	More data → higher mAP ceiling	RECOMMENDED
C	Augment on top of Week 2 baseline	TTA, label smoothing on existing config	LOW RISK

Paper Contribution Strategy

Regardless of Week 4 outcome, the paper's contribution is already strong:

1. **Systematic ablation methodology** — tested 4 configurations with controlled variables

2. **Robustness evaluation framework** — 24-point perturbation sweep (no other steel defect paper does this)
3. **Honest findings** — documented what doesn't work, not just what does
4. **Dataset size analysis** — evidence that 1290 images is insufficient for architectural complexity

Files & References

FILE	PURPOSE
`notebooks/week3_phase_a1_config_fix.ipynb`	A1 training notebook
`notebooks/week3_phase_a2_arch_fix.ipynb`	A2 training notebook
`evals/week3_a1_results.json`	A1 metrics (JSON)
`evals/week3_a2_results.json`	A2 metrics (JSON)
`configs/models/digisteel_v3.yaml`	A2 model config
`digisteel/modules/coord_attention.py`	CoordAttention module